

Draft Paper

From Seasonal Stability to Event-Driven Dynamics: A Spectral Analysis of Groundwater Response to Winter Conditions Near JFK Airport, New York

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Abstract

Long Island depends on a designated sole-source aquifer system, and this groundwater resource is increasingly vulnerable to seasonal road-salt application. This study focuses on the highly urbanized corridor surrounding John F. Kennedy International Airport (JFK), where winter maintenance and impervious surfaces can generate pulses of chloride-rich meltwater during snow events. Rather than directly measuring salt application, this study uses winter temperature conditions as a proxy for likely salting activity. To evaluate how these seasonal conditions relate to groundwater behavior, we analyzed long-term USGS groundwater-level records from wells near JFK. Winters were classified based on the frequency of near-freezing events (-4°C to 1°C), with winters exhibiting one or more such events defined as maximum-salt conditions, and winters with no such events and predominantly warmer temperatures defined as minimum-salt conditions. Spatial and time-series methods, including spectral analysis, periodograms, and wavelet analysis, were used to examine seasonal and multi-year variability. Results show that winters with more frequent near-freezing conditions exhibit reduced low-frequency variability and increased short-term fluctuations, indicating a shift toward a more dynamic and less buffered aquifer response. These findings suggest that winter environmental conditions may alter groundwater system behavior in urban settings. Spectral methods were applied to detect seasonal forcing, multi-year oscillations, and evolving recharge dynamics that cannot be identified solely through linear trend analysis. These methods are used to identify patterns in groundwater response rather than to directly demonstrate the effects of road salt. Differences in spectral power and periodic structure between winter groups are interpreted as potential changes in aquifer response under differing winter conditions. Although the magnitude and specific mechanisms of these effects remain uncertain, the results provide insight into how winter conditions may influence groundwater behavior in urban environments. This work highlights the potential impact of winter conditions on aquifer health and underscores the importance of considering winter maintenance practices in long-term groundwater management.

Introduction

Groundwater is a critical resource for Long Island, which relies on a sole-source aquifer, system for drinking water and ecosystem sustainability (United States Environmental Protection Agency, 2025; New York State Department of Transportation, 2018). In highly urbanized environments such as the region surrounding John F. Kennedy International Airport (JFK), groundwater systems are increasingly influenced by anthropogenic processes, including impervious surface development, stormwater routing, and winter road maintenance practices. Among these, the application of road salt during freeze–thaw conditions is a major driver of freshwater salinization and altered hydrologic behavior (Schlesinger & Bernhardt, 2019; Rodbarry et al., 2023).

Winter conditions introduce highly variable forcing mechanisms to groundwater systems. Freeze–thaw cycles and snowmelt events generate pulses of runoff that infiltrate recharge zones and modify both the chemical and physical behavior of aquifers. While long-term increases in chloride concentrations have been widely documented (Rodbarry et al., 2023; United States Environmental Protection Agency, 2025), less attention has been given to how these seasonal processes influence the temporal structure of groundwater variability across multiple time scales. Unlike many pollutants, chloride behaves conservatively in aquatic systems, meaning it does not readily degrade or transform. As a result, salt introduced during winter months can persist year-round, contributing to sustained increases in background salinity levels (Stroud Water Research Center, 2026; Schlesinger & Bernhardt, 2019).

Long Island’s aquifer system is particularly vulnerable to this type of contamination due to its high permeability, shallow water table, and dependence on local recharge (New York State Department of Transportation, 2018). These characteristics allow surface-derived contaminants to infiltrate rapidly into groundwater. In urban environments, extensive impervious surfaces and stormwater infrastructure further enhance this process by concentrating and redirecting runoff toward recharge areas. Chloride from road salt has been shown to infiltrate drinking water supplies, degrade aquatic ecosystems, and contribute to infrastructure corrosion and mobilization of toxic metals (Rodbarry et al., 2023; Schlesinger & Bernhardt, 2019).

Recent regional assessments highlight the growing scale of this issue. The New York City Department of Environmental Protection (DEP) has documented steadily increasing chloride concentrations across all major water supply systems since the 1980s, with the most rapid increases occurring in the more urbanized East of Hudson watershed (Bureau of Water Supply, 2025). Projections suggest that continued trends could eventually approach or exceed recommended drinking water thresholds, emphasizing the need for proactive management. Although these studies primarily focus on surface reservoirs, they reflect a broader pattern of freshwater salinization occurring across developed regions (Schlesinger & Bernhardt, 2019).

Traditional hydrologic analyses often rely on trend-based or statistical approaches that may overlook non-stationary and multi-scale behavior. However, groundwater systems are

inherently dynamic, with variability driven by interacting processes operating at seasonal, interannual, and short-term scales. Previous work has demonstrated that time series decomposition and signal detection techniques are essential for separating these components and reducing noise in environmental datasets (Parag et al., 2025). Spectral methods such as periodograms and wavelet analysis provide a framework for identifying dominant periodicities and time-frequency variability that cannot be captured through linear analysis alone (Parag et al., 2025).

To address this gap, this study examines groundwater-level responses to winter conditions near JFK Airport using time-series and spectral analysis techniques. Winter conditions are classified using temperature-based proxies for freeze–thaw activity associated with increased road salt application. By comparing groundwater variability between winters with minimal and more frequent freeze conditions, this study aims to determine whether seasonal processes are associated with measurable changes in aquifer response patterns. These findings provide insight into how winter environmental conditions may influence groundwater dynamics in highly urbanized regions and contribute to the broader understanding of freshwater salinization.

Methodology

Groundwater-level and chloride data were retrieved from the U.S. Geological Survey (USGS) Water Data Services using the R package `dataRetrieval` (De Cicco et al., 2023). Chloride concentrations (parameter code 00940, mg/L) and daily groundwater levels (parameter code 62610, depth to water) for 2000–2024 were obtained for wells located within 10 km of John F. Kennedy International Airport (JFK; $-73.7781, 40.6413$). The JFK groundwater is a part of both the Jameco and Upper Glacial aquifers. Groundwater wells were first identified within a bounding box around JFK using the U.S. Geological Survey’s National Water Information System (NWIS) and then clipped to a 10 km buffer. Spatial analyses were conducted in NAD83 / New York Long Island State Plane (EPSG: 2263), and the 10 km buffer was converted from meters to U.S. survey feet to match the projection units. Wells were retained if their coordinates fell within the buffer, and records lacking geographic coordinates were excluded.



Figure 1: Groundwater level time series (2000–2024) from the well located closest to the highest chloride concentration near JFK Airport, with data quality control applied. Observed data are shown in black, short gaps (≤ 14 consecutive days) filled using linear interpolation in blue, and longer gaps left unfilled (red markers). This approach preserves short-term variability while avoiding artificial smoothing that could bias spectral and wavelet analyses.

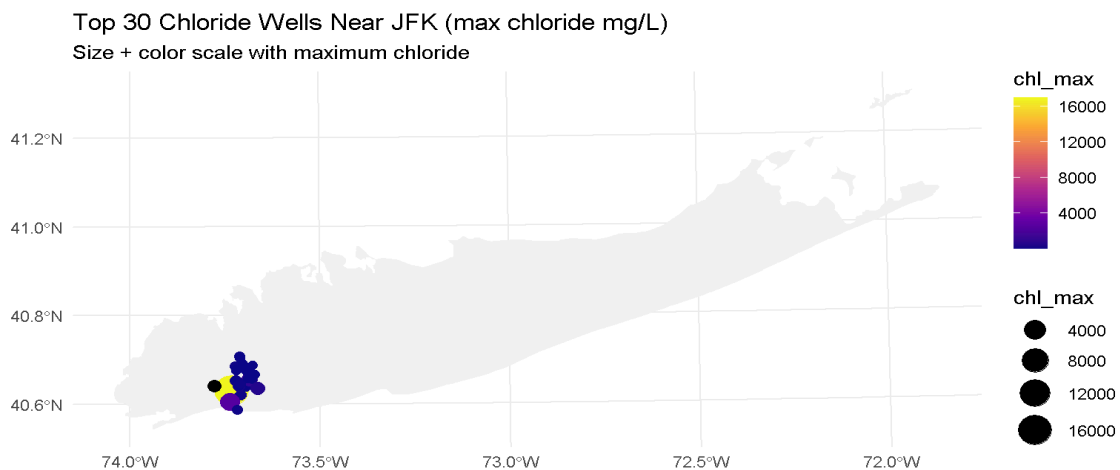


Figure 2: Spatial distribution of the 30 groundwater wells with the highest recorded chloride concentrations near JFK Airport (within 10km radius). Point size and color represent maximum observed chloride concentration (mg/L), with larger and warmer-colored symbols indicating higher values. Wells are clustered within the highly urbanized coastal zone, highlighting areas of elevated salinity and guiding selection of the representative groundwater time series used for spectral analysis.

Groundwater-level records were screened for continuity, as shown in Figure 1. Chloride metadata were used to identify wells with adequate sampling histories, as shown in Figure 2. The groundwater-level well with the longest continuous daily record was selected for detailed spectral analysis. Duplicate dates were removed, and short gaps (≤ 14 days) were linearly interpolated; longer gaps were left unfilled to avoid spectral distortion. To preserve temporal continuity while minimizing distortion of spectral properties, short gaps (≤ 14 consecutive days) were interpolated using a linear method implemented via the `imputeTS` package (`na_interpolation`). This threshold was selected to avoid introducing artificial periodicities, as interpolation across longer gaps can impose smooth transitions that generate spurious low-frequency signals. By restricting interpolation to short intervals, the procedure maintains the integrity of high-frequency variability while preventing bias in spectral and wavelet analyses. December observations were assigned to the following hydrologic winter year. Winters were defined as December–February (DJF) and classified using a temperature-based proxy for likely road-salt application. Daily modeled temperature data (2000–2024) were used to identify near-freezing events (-4°C to 1°C) indicative of freeze/snow maintenance conditions (National Weather Service, n.d.). Near-freezing conditions are strongly associated with snow and ice formation, which typically trigger road salt application in urban environments. Winters with no such events were classified as minimum-salt (min-salt), while winters with one or more events were classified as maximum-salt (max-salt). This classification represents a proxy for relative salting intensity rather than a direct measure of salt application. Daily precipitation data for JFK were obtained from NOAA’s Global Historical Climatology Network (GHCN) via the R package `GHCNr` (National Centers for Environmental Information, n.d.) to provide seasonal context. Precipitation time series were additionally smoothed using moving averages and Kolmogorov–Zurbenko (KZ) filtering to examine seasonal variability. Groundwater data were subset to DJF and analyzed using periodograms and wavelet analysis (Hernandez Gonzalez et al., 2023; Parag et al., 2025). Spectral methods were applied as exploratory tools to identify periodic structure, seasonal forcing, and time-frequency variability in groundwater response, rather than to demonstrate direct causal effects. The DJF series was evaluated both as concatenated seasonal records (exploratory) and as per-winter-averaged power spectral densities to reduce bias arising from artificial adjacency between winters. Comparisons between min-salt and max-salt winters focused on differences in spectral power, dominant periodicities, and frequency structure. Variations in these properties were interpreted as changes in aquifer

response under varying winter conditions, with increased spectral power indicating a stronger or more dynamic groundwater response. These analyses evaluate whether winters with more frequent freeze-and-snow maintenance conditions are associated with changes in groundwater behavior near JFK, rather than establishing road salt as the direct cause. Analyses were conducted in R (version 4.5.3) using the following packages: dataRetrieval, GHCNr, ggplot2, ggspatial, kza, lubridate, purrr, sf, tigris, tidyverse, trend, USAboundaries, viridis, WaveletComp, and zoo. Session details and download dates were saved, and results reflect the USGS data available at the time of retrieval.

Temperature-Based Classification of Winter Salt Regimes

Daily air temperature data from NOAA's *Global Surface Summary of the Day (GSOD)* were used to classify winters based on thermal conditions associated with road-salt application. Winters were categorized as maximum-salt when at least one near-freezing interval (-4°C to 1°C) occurred, and as minimum-salt when such intervals were absent and temperatures were predominantly within a warmer range ($+4^{\circ}\text{C}$ to $+10^{\circ}\text{C}$). This classification follows the temperature-based proxy framework introduced by Gonzalez et al. (2023) for assessing climate-driven processes affecting geomorphological systems, including sinkhole formation.

Methodological Considerations: Seasonal Segmentation and Spectral Integrity

A key methodological consideration in this study is the treatment of discontinuous seasonal data in spectral and wavelet analyses. When DJF segments from multiple years are concatenated into a single time series, artificial adjacency is introduced between the end of one winter and the beginning of the next. This creates discontinuities that do not represent real hydrologic processes and can introduce broadband spectral energy, potentially distorting periodograms, wavelet power, and the interpretation of dominant timescales. To avoid these effects, winters are analyzed independently, preventing the mixing of temporally unrelated observations. In the present analysis, concatenated DJF series were used only for exploratory visualization, while the primary results rely on per-winter spectral estimation and subsequent averaging across winters. By computing multitaper spectra and wavelet power independently for each winter and then aggregating these estimates, we reduce bias associated with artificial boundary effects and preserve the intrinsic temporal structure of each season. This approach ensures that detected periodicities reflect within-winter groundwater dynamics rather than artifacts of data construction, thereby strengthening the robustness of the inferred differences between minimum-salt and maximum-salt winter conditions.

Results

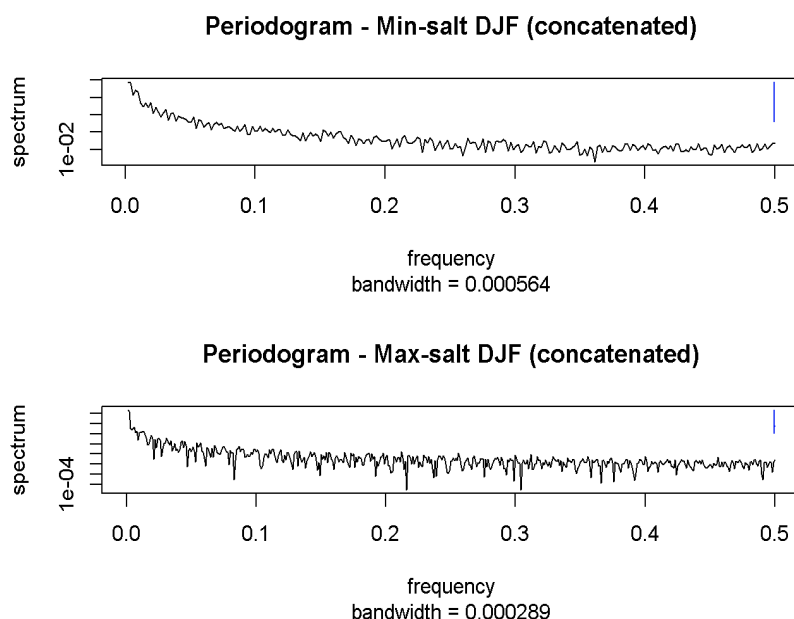


Figure 3. Periodogram analysis of groundwater levels during DJF periods showing differences in frequency structure between minimum-salt and maximum-salt winters.

Periodogram analysis of groundwater levels during DJF periods reveals a clear difference in frequency structure between minimum-salt and maximum-salt winters. As shown in Figure 3, minimum-salt winters exhibit spectral energy concentrated at low frequencies, indicating dominance of seasonal-scale variability and a relatively smooth, well-filtered groundwater response. In contrast, maximum-salt winters exhibit reduced overall spectral power and a more irregular distribution of energy across frequencies, including increased variability in the mid- and high-frequency ranges. This pattern suggests a shift from a stable, low-frequency-dominated system to a more complex, less filtered response under presumed high-salt conditions. The reduction in spectral coherence and increase in higher-frequency variability indicate that groundwater dynamics become more irregular during winters associated with increased freeze-thaw activity and snow maintenance practices, conditions associated with increased road salt application.

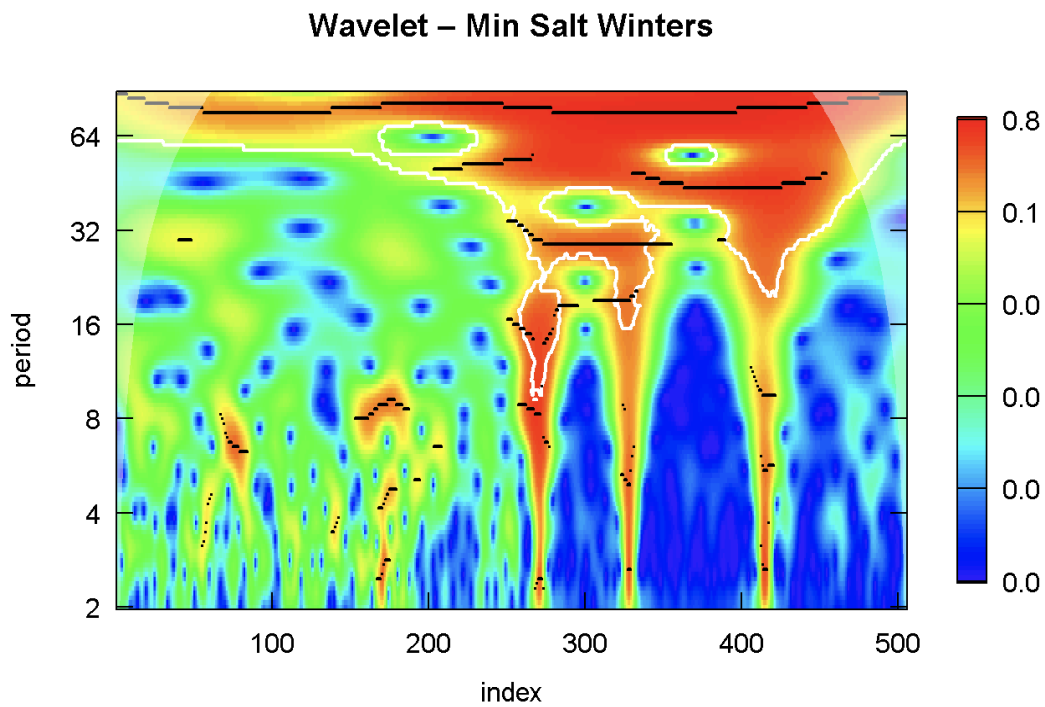


Figure 4. Wavelet power spectrum of groundwater levels during minimum-salt winters, showing dominant spectral energy at longer periods (~30–90 days) and a continuous band indicative of stable, seasonally driven variability.

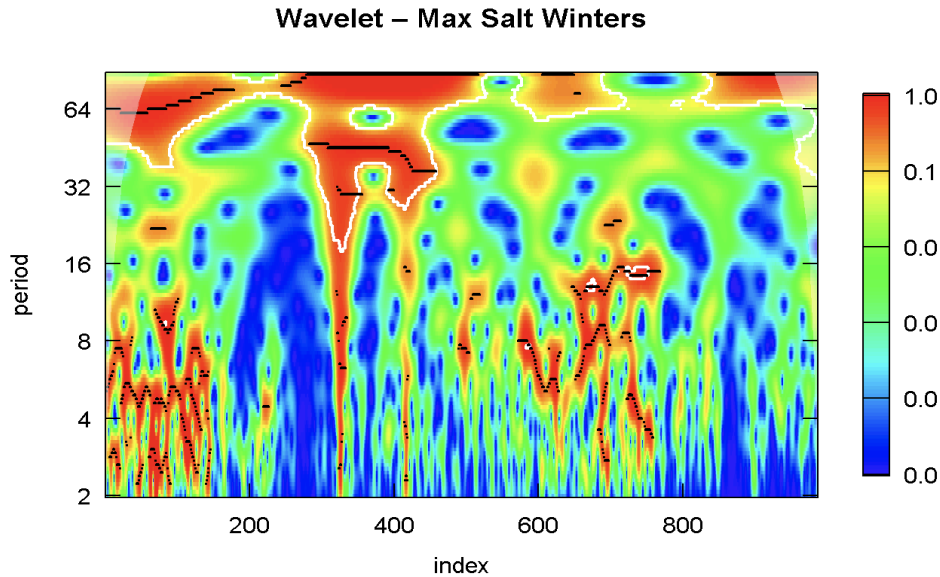


Figure 5. Wavelet power spectrum of groundwater levels during maximum-salt winters, showing a broader and more fragmented distribution of spectral power across both long and short periods, indicating increased short-term variability and reduced temporal coherence.

As shown in Figures 4 and 5, the color gradients represent wavelet plots of spectral power as a function of time and period, with warmer colors (red and orange) indicating higher power and cooler colors (blue) indicating lower power. In the minimum-salt winters, high-power regions are concentrated in the upper portion of the plots (longer periods), forming continuous bands that reflect persistent seasonal signals. In contrast, maximum-salt winters exhibit high-power regions that are more widely distributed across both long and short periods, with frequent instances of elevated power at shorter timescales. This broader and more irregular distribution indicates that variability is both stronger and less temporally consistent. The presence of intermittent high-power regions at short periods further suggests increased influence of short-term fluctuations, reinforcing the interpretation that aquifer response becomes more variable and less filtered under high-salt conditions.

Wavelet analysis reveals clear differences in the temporal structure of groundwater variability between minimum-salt and maximum-salt winters. Minimum-salt winters exhibit a strong, continuous band of spectral power at longer periods (~30–90 days), indicating the dominance of stable, seasonal-scale groundwater fluctuations. This pattern remains consistent through time, reflecting a relatively coherent and predictable recharge regime. In contrast, maximum-salt winters display a more fragmented distribution of spectral power, with increased energy at shorter periods (2–16 days) and reduced continuity of the seasonal band. These patterns indicate that groundwater response becomes less stable and more irregular under high-salt

conditions, reflecting a shift from a consistent, seasonally driven system toward a more dynamically variable and less filtered aquifer response during winters associated with increased freeze–thaw activity.

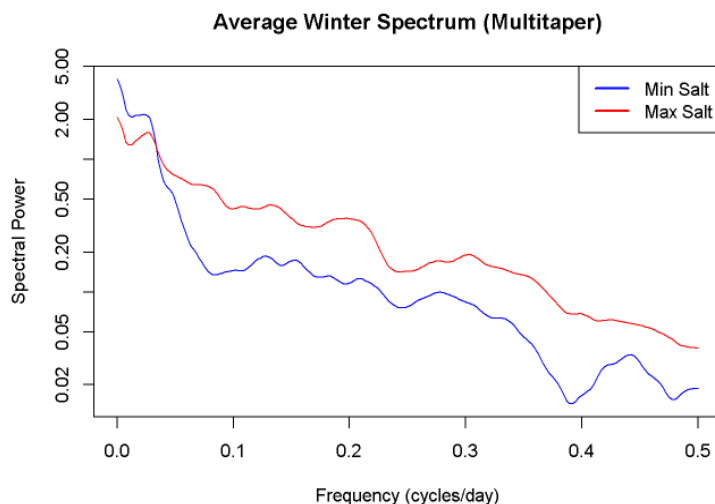


Figure 6. Frequency-domain spectral comparison of groundwater levels during minimum-salt and maximum-salt winters, showing reduced low-frequency power and a flatter spectral distribution under maximum-salt conditions.

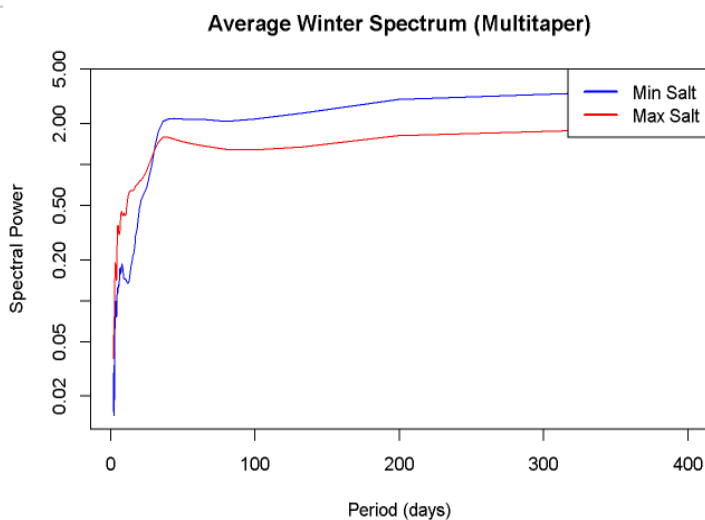


Figure 7. Period-domain spectral comparison of groundwater levels, illustrating reduced spectral power at longer periods (~100–300 days) during maximum-salt winters, indicating diminished seasonal variability.

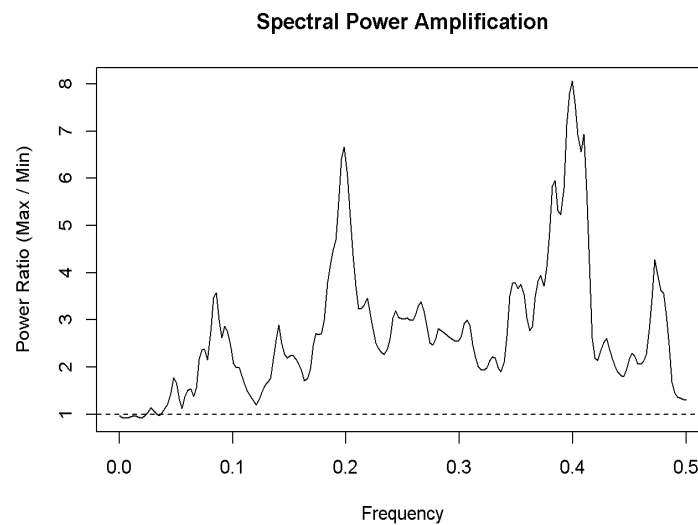


Figure 8. Spectral power ratio (maximum-salt divided by minimum-salt) across frequencies, showing amplification of groundwater variability under maximum-salt conditions, particularly at higher frequencies.

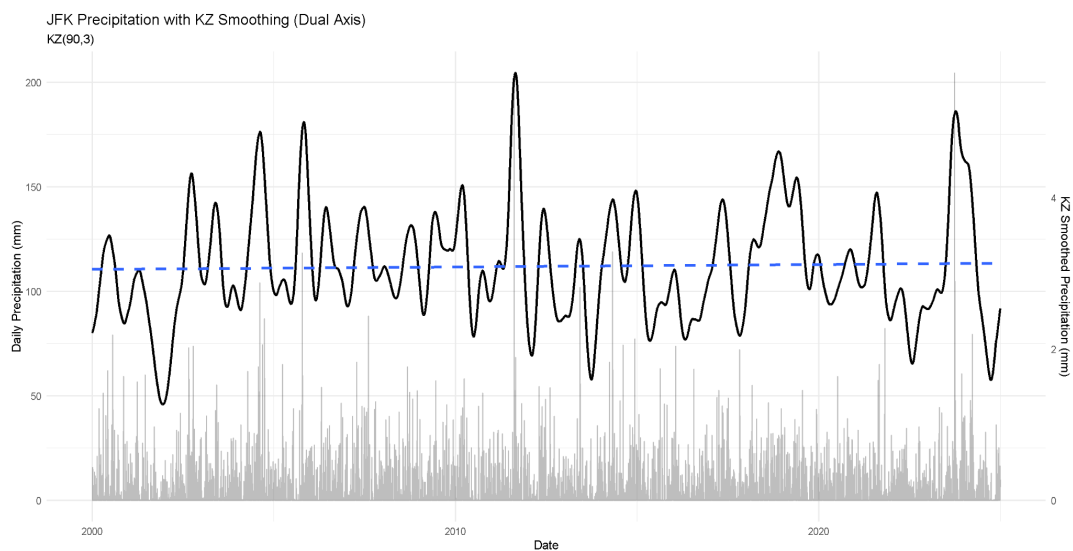


Figure 9. Daily precipitation at JFK (2000–2024) shown as gray bars, with a Kolmogorov–Zurbenko (KZ) smoothed series ($m = 90$, $k = 3$) overlaid as a black line (right axis).

As shown in Figures 7 and 8, spectral comparisons in both the period and frequency domains further highlight differences between minimum-salt and maximum-salt conditions. The frequency-domain spectral comparison shows clear differences in groundwater variability between minimum-salt and maximum-salt winters. Minimum-salt winters exhibit higher spectral power at low frequencies, indicating a strong dominance of long-timescale (seasonal) groundwater fluctuations. In contrast, maximum-salt winters display reduced power in this low-frequency range and a comparatively flatter spectral distribution, suggesting that the seasonal signal is less pronounced under high-salt conditions. At higher frequencies, representing short-term variability, the differences between the two groups are smaller, indicating that short-period fluctuations remain present in both cases. Overall, the reduced low-frequency power in maximum-salt winters indicates a weakening of structured, seasonal groundwater dynamics.

The period-domain spectral comparison further highlights the attenuation of seasonal groundwater variability during maximum-salt winters. Minimum-salt winters show greater spectral power at longer periods (~100–300 days), corresponding to strong and coherent seasonal recharge signals. In contrast, maximum-salt winters exhibit consistently lower spectral power across these longer periods, indicating a diminished seasonal component. Differences between the two groups are less pronounced at shorter periods, suggesting that short-term variability is relatively similar between conditions. This pattern confirms that the primary distinction between winter types lies in the strength of long-period variability, with maximum-salt winters characterized by a reduction in dominant seasonal groundwater behavior.

To assess statistical significance, a Welch two-sample t-test was conducted to compare spectral power between minimum-salt and maximum-salt winters, using log-transformed spectral estimates. The results indicate a statistically significant difference between the two groups ($t = -5.70$, $df = 383.62$, $p < 0.001$). The average spectral power for maximum-salt winters was higher than for minimum-salt winters, particularly at higher frequencies, with the 95% confidence interval for the difference ranging from -1.21 to -0.59, indicating consistently greater variability under presumed higher-salt conditions. These results confirm that the observed differences in spectral structure are not due to random variation, but instead reflect a systematic shift in groundwater dynamics associated with winter conditions characterized by increased freeze–thaw activity. The statistical significance of this result supports the interpretation that groundwater systems exhibit enhanced and more variable responses during maximum-salt winters.

The spectral power ratio (maximum-salt divided by minimum-salt) shown in Figure 8 shows consistent amplification of groundwater variability under presumed high-salt conditions

across the entire frequency spectrum. Values greater than 1 across nearly all frequencies indicate that maximum-salt winters exhibit greater variability than minimum-salt winters at both seasonal and short timescales. Notably, amplification is particularly pronounced at higher frequencies, where power ratios range from 5 to 8, indicating strong enhancement of short-period fluctuations. This pattern demonstrates that the influence of winter conditions associated with frequent freeze–thaw events is not limited to a specific frequency band but instead reflects a broad increase in groundwater variability. These findings reinforce previous spectral and wavelet analyses, confirming that groundwater systems under high-salt conditions exhibit more dynamic and less filtered behavior.

As shown in Figure 9, analysis of daily precipitation at JFK indicates no significant long-term trend in precipitation, despite short-term variability in precipitation, providing hydrologic context for groundwater variability analyses, characterized by frequent fluctuations and intermittent high-magnitude events. Notably, extreme precipitation events, such as those associated with Hurricane Sandy, appear as pronounced peaks in the record, highlighting the variable nature of intense rainfall in the region. To isolate longer-term patterns from this high-frequency noise, a Kolmogorov–Zurbenko (KZ) filter (90-day window, 3 iterations) was applied. This smoothing technique effectively removes short-term variability while preserving seasonal-to-interannual structure, allowing for clearer interpretation of underlying trends. The resulting smoothed precipitation series remains relatively stable over the study period, fluctuating around a consistent mean without exhibiting a sustained directional trend. Despite the presence of extreme events, overall precipitation appears to be mean-reverting. The combination of high short-term variability and long-term stability suggests that observed changes in groundwater dynamics are unlikely to be driven by changes in precipitation input, but instead reflect variations in aquifer response behavior.

Discussion

The results of this study indicate that groundwater behavior near JFK Airport differs between winters with minimal freeze–thaw activity and those with more frequent events. Spectral and wavelet analyses show that winters with more frequent near-freezing events exhibit greater variability across seasonal and shorter timescales, suggesting a shift toward a more dynamic, less filtered groundwater system. In contrast, winters with fewer freeze events tend to exhibit more stable, low-frequency-dominated behavior consistent with a buffered aquifer response. These patterns are consistent with previous research showing that winter processes, including those associated with road salt application, can influence groundwater systems by increasing chloride inputs and altering subsurface flow dynamics. However, because this study uses temperature as a proxy rather than direct salt measurements, the results should be interpreted as reflecting broader winter environmental effects rather than road salt application

alone. Other factors, such as freeze–thaw impacts on soil structure, precipitation variability, and urban runoff, may also contribute to the observed patterns. Additionally, precipitation was considered but not emphasized in the main analysis for clarity. Precipitation is an important factor influencing both road salt application and groundwater recharge, and may therefore affect the study in multiple ways. However, precipitation data in Figure 9 show no significant long-term variability in precipitation over the course of a 24-year period sufficient to influence aquifer recharge, as indicated by the relatively stable long-term average. Barring some exceptions, precipitation data indicates peaks balanced by corresponding low periods that average out over time, reducing its influence on the observed groundwater variability. Due to precipitation having an adverse effect on the amount of salt applied during maximum salt times, precipitation was still recorded and specific periods were analyzed, but found to have a negligible influence on the results. Cross-referencing temperature and precipitation could introduce additional complexity in a study that already uses temperature as a proxy, with the expectation that there are several external factors affecting aquifer recharge conditions. Overall, these findings suggest that winter conditions play an important role in shaping groundwater variability in urban coastal aquifers and highlight the need to consider seasonal processes in groundwater management.

Study Area: JFK as a Road Salt Case Study

This study focuses on a localized area surrounding JFK in Jamaica, Queens, rather than the broader NYC metropolitan region. New York City derives approximately 90% of its potable water from major reservoir systems; the latter half consisting of lakes, rivers, The New Croton Reservoir and Queen’s groundwater supply system in the southeastern region subsisting the 10%. The Catskills and Delaware watersheds make a majority of NYC’s drinking water through reservoir collection and distribution (National Center for Biotechnology Information, 2021). The Croton watershed, located north of New York City and separate from the Long Island aquifer system, has experienced salinity increases due to geological factors, local wastewater treatment plants, fractured bedrock and glacial soils, slow dispersal of contamination, and as well as suburban activity, which may limit its suitability for isolating urban-scale effects compared to JFK. In 2021, the New York City Department of Environmental Protection had tasked an internal Salinity Task Force in order to better understand salinity in the NYC watersheds through its analysis stemming as far as 1985 in upstate’s reservoirs. Following their investigation the task force found a sustained increase in chloride concentrations in all NYC’s reservoirs. The West of Hudson watersheds have relatively low levels of chloride, compared to the East of Hudson’s watersheds, with the highest increases recorded in The Croton watershed (NYC Department of Environmental Protection, 2025). While the Jamaica, Queens groundwater system contributes less than 10% of NYC’s potable water, the suburban and city implications to salt application represent a far larger trend followed by cities in snowbelts. JFK’s local wells and aquifers

concisely isolate numerous factors plaguing larger regions of study regarding road salt's increased freshwater salinity, including anthropogenic weathering, wastewater inputs, urban construction, fertilizer use, resource extraction, water softening, saltwater intrusion, and climate change. While JFK is not limited to road salt intrusion and subjected to salinity increases via external factors, it provides a unique advantage as a study area due to its concentrated and consistent winter salt application within its large surface area characterized by intensive salt application during winter conditions.

Interpretation of Spectral Power and Timescales in Groundwater Dynamics

Spectral analysis helps interpret groundwater dynamics across multiple timescales by decomposing variability into frequency components. In this context, low frequencies (long periods) represent seasonal to multi-month groundwater behavior associated with recharge, storage, and delayed release processes, while high frequencies (short periods) reflect rapid responses to individual events such as precipitation pulses, snowmelt, or transient surface inputs.

Spectral power (or spectral density) quantifies the strength of variability at each timescale. Higher spectral power indicates that fluctuations at a given period are more pronounced and consistent, meaning that the corresponding process strongly influences groundwater behavior. In hydrologic terms, strong low-frequency power reflects a coherent and efficient aquifer system, where recharge is stored and released gradually, producing stable seasonal signals. In contrast, elevated high-frequency power indicates a system that responds rapidly to short-term forcing, suggesting reduced buffering capacity and more event-driven dynamics.

In this study, minimum-salt winters exhibit stronger spectral power at longer periods, indicating a dominant seasonal recharge signal and a more stable, storage-controlled aquifer response. Conversely, maximum-salt winters show increased power at shorter periods and reduced coherence at longer periods, reflecting a shift toward more variable and less filtered groundwater dynamics. This redistribution of spectral energy suggests that aquifer behavior transitions from a regulated, storage-dominated system to a more reactive system influenced by short-term inputs under salt-intensive winter conditions.

It is important to note that a transition in behavior is observed around ~30–40 days, representing a characteristic timescale separating short-term event-driven responses from longer-term storage processes. This crossover provides insight into the effective “memory” of the aquifer system, beyond which groundwater dynamics become dominated by seasonal recharge rather than transient inputs. This transition is consistent with the observed crossover in spectral power

between minimum-salt and maximum-salt winters, where short-period variability becomes dominant under high-salt conditions while long-period structure weakens (Figure 7).

Conclusion

This study examined how winter conditions associated with freeze–thaw activity influence groundwater behavior near JFK Airport using spectral, time–frequency, and statistical analyses. Across multiple methods, consistent differences were observed between minimum-salt and maximum-salt winters, indicating that winter conditions are associated with measurable changes in aquifer response dynamics. Spectral analyses, including periodograms and multitaper spectral estimates, show that minimum-salt winters are characterized by stronger low-frequency (seasonal-scale) groundwater variability, reflecting a stable and well-structured recharge regime. In contrast, maximum-salt winters exhibit reduced spectral power at these low frequencies and a comparatively flatter spectral distribution. This indicates a weakening of the dominant seasonal groundwater signal rather than a simple increase in overall variability. Period-domain representations further support this interpretation, showing diminished power at longer periods (~90 days) during maximum-salt winters, consistent with attenuation of coherent seasonal recharge processes. Wavelet analysis reinforces these findings by demonstrating that minimum-salt winters maintain continuous, high-power bands at longer periods, indicative of stable seasonal forcing. In contrast, maximum-salt winters display more fragmented and temporally inconsistent patterns, with intermittent high-power regions across both long and short timescales. This suggests that groundwater dynamics under these conditions become less temporally coherent and more irregular.

Statistical analysis confirms that these differences are significant. The Welch two-sample t-test indicates a clear separation in spectral characteristics between winter types, demonstrating that the observed changes in groundwater variability are systematic rather than random. Additionally, spectral power ratio analysis shows that while short-term variability may be amplified in maximum-salt winters, this occurs alongside a reduction in structured seasonal behavior, indicating a shift in how variability is distributed across timescales rather than a uniform increase in system energy. Importantly, precipitation analysis provides critical context for interpreting these results. Despite substantial short-term variability and episodic extreme events, including large precipitation peaks associated with storms such as Hurricane Sandy, the long-term smoothed precipitation record (via KZ filtering) remains stable and does not exhibit a sustained trend. This indicates that precipitation has remained relatively constant over the study period. As a result, the observed changes in groundwater behavior are unlikely to be driven directly by changes in rainfall input, but instead reflect alterations in aquifer response dynamics.

Overall, the consistency across multiple analytical approaches (periodograms, wavelet transforms, spectral comparisons, and statistical testing) indicates that the observed differences

are robust. These findings suggest that winter conditions associated with frequent freeze–thaw activity are linked to measurable changes in groundwater behavior, characterized by increased variability and altered frequency structure. While this study does not directly confirm these changes to road salt application, the results are consistent with the hypothesis that environmental conditions associated with salt-intensive winters (such as sodium chloride application) may influence aquifer response dynamics in the JFK Upper Glacial and Jameco aquifers. A suggestion for future research would be to access data of actual existing NaCl road salt application (instead of proxy data) times in the Jameco and Upper glacial aquifer of the target region.

Credit Authorship Contribution Statement

Neville, J: Editing, writing- Results, Conclusion, Probatory R-studio Coding-figure 7, unused CCF lag graphs, research groundwork, R-Studio reviewing; Marsellos, A.E.: supervision, Rstudio coding, guidance, editing, Discussion; Tsakiri, K.G.: Rstudio coding review; Mighty, M: peer researcher, research groundwork, writing- discussion, editing; Bernhardt, L: writing - abstract and introduction, research, Rstudio coding, revising.

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dataRetrieval

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zoo

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Versions of Draft Paper (Past)

Potential Titles

- Chemical Enhancement of Percolation by Salts and Its Role in Aquifer Recharge and Sinkhole Development
- How Winter Road Salt Shapes Long Island's Groundwater
- Seasonal Road-Salt Impacts on Long Island's Aquifer System
- Tracking Road-Salt Pulses in Long Island Groundwater
- Winter Salt Use and Changing Groundwater Patterns on Long Island
- Seasonal Salt Forcing and Hydrogeologic Vulnerability on Long Island
- Winter Salt as a Growing Pressure on Long Island's Sole-Source Aquifer
- Road-Salt Forcing and Early Signs of Groundwater Change on Long Island
- Winter Salt Signals in Long Island's Aquifer
- Tracking Salt in Long Island Groundwater

Abstract

Draft Abstract - Version 1:

Road salt is one of the most widespread winter contaminants in cold-region environments, and its increasing use has created a measurable geophysical and hydrogeochemical signal across soils, surface waters, and groundwater systems. This project examines how road-salt-derived chloride and associated pollutants move through snow, meltwater, and subsurface pathways by comparing physical and chemical properties of snow collected from urban and rural sites on Long Island. To evaluate how winter-storm patterns contribute to groundwater salinization, we use spectral techniques, including wavelet analysis and periodograms, to estimate the timing and intensity of high-snowfall periods and to correlate these intervals with increases in chloride and related salinity indicators in the subsurface. Based on previous research demonstrating that road salt is a major driver of rising groundwater salinity in northeastern watersheds, the project also examines how salt alters soil chemistry and infiltration dynamics. Field and literature-based evidence show that salt can temporarily enhance infiltration by compressing clay particle spacing, yet sodium-rich salts ultimately degrade soil structure, reduce hydraulic conductivity, and promote preferential flow channels that allow contaminants to bypass natural filtration. These mechanisms suggest that pollutant loads will be higher in urban snow and that repeated winter storms contribute to cumulative chloride buildup in groundwater. While the precise magnitude of these effects on Long Island remains uncertain, the combined snow-chemistry data, soil-infiltration processes, and spectral trends are expected to reveal emerging patterns of chloride accumulation and altered recharge behavior. The discussion explores these patterns as emerging signals rather than definitive causal conclusions, acknowledging the complexity of subsurface transport and the limits of short-term sampling. Overall, the study highlights the scientific and social implications of continued road-salt use, including risks to aquifer health,

infrastructure corrosion, and ecosystem stress, and underscores the need for improved winter-maintenance strategies that balance road safety with long-term groundwater protection.

Revised - Abstract - Version 2:

Long Island depends on a sole-source aquifer system, and this groundwater is becoming more at risk from seasonal road-salt use. This seasonal anthropogenic salt forcing creates a clear hydrogeologic vulnerability for the region. When snowfall is followed by two or more days below 33°F, road crews apply salt repeatedly, creating strong pulses of chloride-rich meltwater that move into the ground. This project studies how these seasonal salt pulses affect groundwater by combining snow-chemistry data with long-term USGS groundwater-level and conductance records. Snow chemistry shows how much salt enters the system during storms, while the groundwater data show how the aquifer responds over time. To track these responses, we used spectral tools, including wavelet analysis and periodograms. Spectral methods are applied to detect seasonal forcing, multi-year oscillations, and evolving recharge dynamics that cannot be identified solely through linear trend analysis. Salt can temporarily increase infiltration by tightening clay spacing, potentially amplifying seasonal groundwater fluctuations. Over time, sodium can damage soil structure, slow recharge, and create fast pathways that move salt deeper into the aquifer. These processes suggest higher pollutant loads in urban snow and gradual chloride buildup after repeated winter storms. Although the exact size of these effects on Long Island is still uncertain, the combined snow-chemistry data, soil-infiltration processes, and spectral trends reveal early signs of changing recharge and rising salinity. This work highlights the growing impact of road salt on aquifer health and the need for winter-maintenance practices that protect both safety and long-term water quality.

Outline/Ideas:

Introduction

Road salt is one of the most widespread winter contaminants in cold-region environments, and its increasing use has created a measurable geophysical and hydrogeochemical signal across soils, surface waters, and groundwater systems. This project examines how road-salt–derived chloride and associated pollutants move through snow, meltwater, and subsurface pathways by comparing physical and chemical properties of snow collected from urban and rural sites on Long Island.

Methodology

First Version:

In addition to analyzing chloride concentration, electrical conductivity, pH, total dissolved solids, and nutrient indicators, the study evaluates how salt alters soil chemistry and infiltration dynamics through the process known as chemical enhancement of percolation. Salt can initially increase infiltration by compressing clay particle spacing and opening pore pathways, but sodium-rich salts ultimately degrade soil structure, clog pores, reduce hydraulic conductivity, and create preferential flow channels that allow contaminants to bypass natural filtration. These changes can temporarily increase aquifer recharge but often reduce long-term recharge efficiency while degrading groundwater quality through elevated salinity, mobilized metals, and higher TDS. Spectral data will be used as well using wavelets and periodograms to show the impact of road salt on groundwater over a period of time. Integrating primary snow-sample data with regional salinity trends, spectral analysis, and freshwater-salinization research, the project assesses the scale of road-salt impacts on aquifers, ecosystems, and infrastructure. The expected findings include significantly higher pollutant loads in urban snow and clear evidence that repeated winter storms and heavy salt application contribute to long-term chloride accumulation in groundwater. Ultimately, the study underscores the need for improved winter-maintenance practices and explores alternative de-icing strategies, such as brine, porous pavement, and reduced-salt management, to protect aquifer health while maintaining road safety.

Discussion/literature review

[-Chloride hydrogeochemistry of the finger lakes in Central and Western New York, USA - ScienceDirect](#)

Since New Hampshire adopted the use of road salts as policy in 1938 many states in snowy weather followed suit. The use of road salts became widespread due to its ability to lower the freezing point of water.

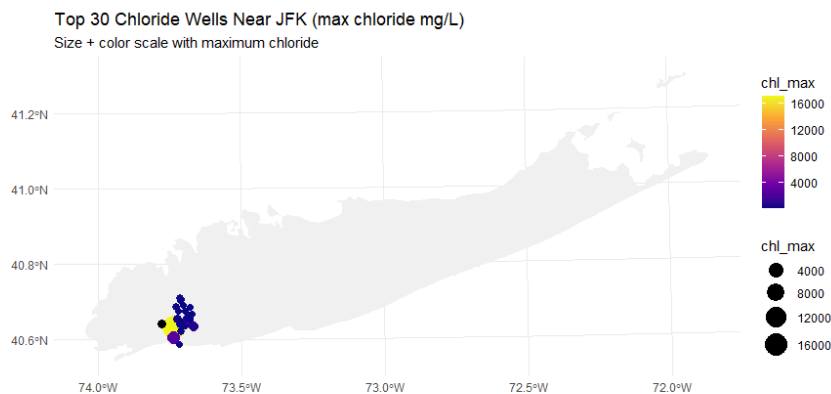
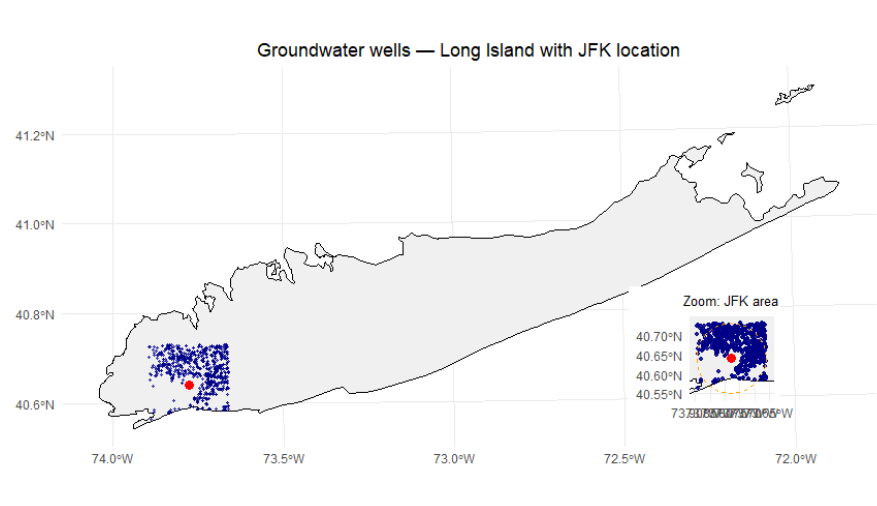
Some important questions for our data:

“Minimum-salt winter times vs maximum-salt winter times”. How are we going to select which winters are considered minimum or maximum-salt affected, so to collect the data and scan for different aquifer behavior (frequency of replenishment)

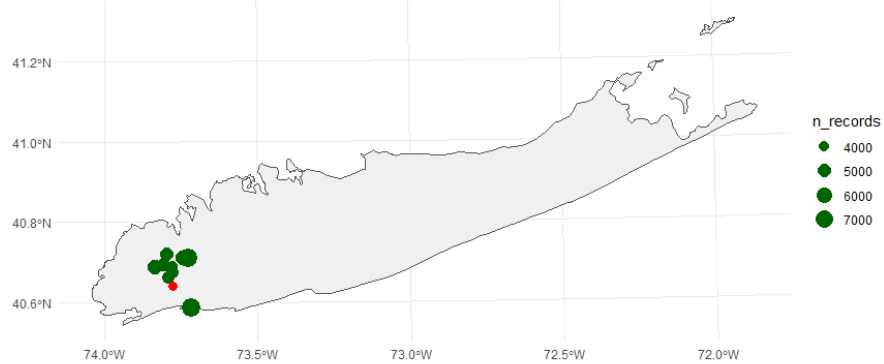
What triggers salt distribution on the roads who decides and with what criteria

<https://www.nyc.gov/assets/dep/downloads/pdf/water/drinking-water/salinity-management-assessment.pdf>

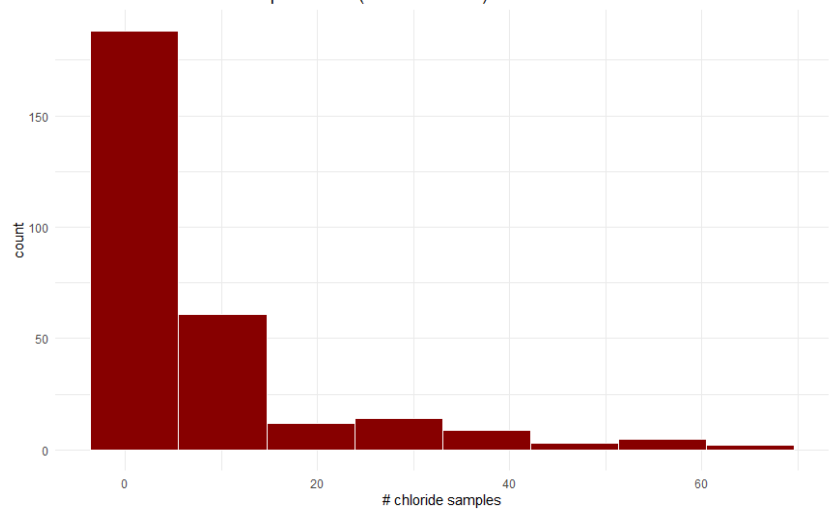
- In 2021 the NYC Department of Environmental Protection (DEP) commissioned an internal Salinity Task Force to examine, measure, and understand salinity in the NYC watersheds and water supply, in order to mitigate and monitor salinity across the cities history 1) Leading cause of groundwater increase salinity is salt contamination



Top 10 Groundwater-Level Wells Near JFK (by Record Count)



Distribution of Chloride Sample Counts (Wells near JFK)



Data

Here is some general feedback and suggestions based on your current draft abstract:

The current draft demonstrates strong understanding of road salt hydrogeochemistry and environmental relevance. The mechanisms described are scientifically accurate and thoughtfully presented.

However, the project focus must be sharpened.

At present, the narrative emphasizes snow chemistry and pollutant comparison between urban and rural sites. While this is valuable, it does not yet fully integrate the hydrogeologic time-series framework that defines the core scientific question.

The primary objective should be:

To understand how seasonal anthropogenic salt forcing modifies the temporal structure and long-term behavior of Long Island groundwater systems.

Spectral methods (periodogram and wavelet analysis) must not appear as advanced add-ons. They must be justified because:

- Road salt application is strongly seasonal.
- Recharge cycles are seasonal.
- Long Island aquifers exhibit oscillatory behavior.
- The system is non-stationary over decades.

The abstract should clearly communicate that spectral tools are necessary to detect evolving frequency structure, seasonal amplification, and possible cumulative hydrogeochemical modification.

Additionally, the infiltration discussion should connect directly to measurable signals. For example:

If salt enhances infiltration temporarily, do we observe increasing seasonal amplitude in groundwater response?

If sodium degrades soil structure over time, do we observe altered recharge damping or increasing variability?

The work should move from:

Describing mechanisms

to

Testing whether time-series evidence reflects those mechanisms.

The strongest version of this project positions groundwater dynamics at the center, with snow chemistry serving as supporting evidence of seasonal forcing.

Specific Improvements Recommended

1 Shift the Opening Focus

Instead of beginning with snow sampling, begin with:

- Long Island's sole-source aquifer system
- Seasonal anthropogenic salt forcing
- Hydrogeologic vulnerability

Then introduce snow chemistry as evidence of input magnitude.

2 Strengthen Spectral Justification

Add one sentence explaining:

Spectral methods are applied to detect seasonal forcing, multi-year oscillations, and evolving recharge dynamics that cannot be identified through linear trend analysis alone.

That sentence alone elevates scientific clarity.

3 Clarify Data Integration

Explicitly state:

- USGS groundwater level time series are analyzed.

- Chloride or conductance records are compared.
- Snow chemistry supports interpretation of seasonal input intensity.

Right now that integration is implied but not clearly stated.

Script we need:

We need the combination of snowfall and continuous below 33F temperatures for 2 or more consecutive days

Terminology change?

Instead of calling snow-free and snow-winter times, to call them minimum-salt and maximum-salt winters.

READINGS

32nd Conference on the Geology of Long Island and Metropolitan New York, April 2025

**Wavelet Analysis of Groundwater Level Fluctuations
in Long Island and The Metropolitan Area, NY**

[https://www.stonybrook.edu/commcms/geosciences/about/_LIG-Past-Conference-abstract-pdfs/
2025-Abstracts/1_Parag.pdf](https://www.stonybrook.edu/commcms/geosciences/about/_LIG-Past-Conference-abstract-pdfs/2025-Abstracts/1_Parag.pdf)

Class time version

Title: see at the end potential titles

Road deicing practices introduce substantial chloride loads into shallow aquifers, potentially altering groundwater recharge dynamics over short and long timescales. While short-term salinity increases may enhance permeability by widening pore spaces and accelerating percolation, prolonged exposure may degrade aquifer structure and reduce long-term recharge efficiency. This study investigates the impact of minimum-salt (warmer) versus maximum-salt (freezing) winters on groundwater level fluctuations in Long Island, New York, with a focused analysis on the JFK well station.

The JFK station was selected based on a comprehensive chloride contamination reconstruction conducted using RStudio and publicly available USGS water-quality records from 1930–1999. GIS-based spatial interpolation across Long Island, using the top 30 most contaminated wells, demonstrates that the JFK region consistently exhibits some of the highest chloride concentrations, exceeding 1000 mg/L in multiple observations. The site's flat topography, dense transportation infrastructure, airport runways, parking facilities, and high-risk road networks make it particularly vulnerable to repeated and heavy salt application, establishing it as an ideal natural laboratory for studying long-term aquifer contamination effects.

Freezing versus warmer winters will be identified using classifications from Gonzalez et al. (2024), originally developed in their analysis of sinkhole formation in Long Island. Groundwater level records from the JFK station will be filtered for December–January–February (DJF) during these classified years. Spectral methods, including periodogram and wavelet analyses, will be applied to detect differences in fluctuation frequency, spectral power, and temporal evolution between high-salt and low-salt winters.

We hypothesize that maximum-salt winters may initially show enhanced short-period recharge signals due to temporary increases in permeability, while repeated high-salt exposure over decades may result in dampened seasonal recharge frequencies and reduced spectral energy, reflecting declining aquifer replenishment efficiency. Detecting systematic shifts in dominant frequencies or spectral power will provide quantitative evidence of whether chronic salt loading modifies aquifer dynamics in coastal northeastern hydrogeologic systems.

Updated Research Plan (Key Additions)

1. Study Area Justification: JFK Well Station

1.1 Chloride Contamination Reconstruction

We previously developed a GIS-based chloride contamination assessment for Long Island using:

- USGS publicly available water-quality data (1930–1999)
- The top 30 most chloride-contaminated wells across Long Island
- Spatial interpolation and mapping conducted in **RStudio**
- Time-series chloride reconstruction

Results show:

- The **JFK well station consistently ranks among the most contaminated locations**
- Chloride concentrations exceeding **1000 mg/L**
- Persistent elevated chloride over decades

This makes JFK:

- A confirmed long-term high-salt exposure site
- A statistically defensible choice for focused spectral analysis

1.2 Physical Vulnerability of the JFK Region

The region is characterized by:

- Extremely flat topography
- High-density transportation corridors
- Airport runways and taxiways
- Extensive parking infrastructure
- Rental car facilities

- Heavy winter deicing operations

These factors create:

- High salt application intensity
- High infiltration potential
- Repeated seasonal salt loading
- Chronic chloride exposure

Thus, the JFK well serves as an ideal case study for assessing long-term salt-induced aquifer behavior modification.

2. Refined Study Objectives

1. Focus exclusively on the **JFK well groundwater level time series**.
 2. Classify winters into:
 - Maximum-salt (freezing winters)
 - Minimum-salt (warmer winters)
(Based on Gonzalez et al., 2024.)
 3. Extract DJF groundwater levels for those classified years.
 4. Compare spectral signatures between winter types.
 5. Evaluate long-term trends in dominant frequency and spectral power.
-

3. Why a Single-Well Deep Analysis Is Scientifically Justified

Instead of spatial averaging across multiple wells:

- JFK represents a **controlled high-impact system**.
- Long contamination history improves signal detection.
- Reduces spatial heterogeneity noise.
- Allows deeper frequency-domain interpretation.

This aligns with your signal detection expertise — maximizing signal-to-noise ratio in a high-forcing environment.

4. Additional Hypothesis Strengthening

Given chloride levels >1000 mg/L:

- Aquifer matrix alteration is likely non-negligible.
- Long-term structural changes may already be observable.
- If degradation exists, it should be strongest at JFK.

If no degradation trend is detected even at JFK:

- That suggests resilience of the aquifer system.

Either outcome is scientifically meaningful.

5. Expected Contribution

This study will:

- Be the first to combine:
 - Long-term chloride GIS reconstruction
 - Winter classification from sinkhole climatology

- Spectral analysis of groundwater fluctuations
 - Provide frequency-domain indicators of aquifer health
 - Establish JFK as a sentinel site for salt impact studies
-

If you'd like, I can now:

- Turn this into a **high-impact journal-style abstract (Water Resources Research level)**
- Or compress it into a **300-word AGU abstract**
- Or expand it into a **grant-ready 3–4 page proposal**
- Or integrate equations and spectral metrics formally (since this fits your signal detection expertise very well)**



High-Impact Journal Style (Technical & Strong)

1. **Spectral Detection of Aquifer Recharge Efficiency Under Chronic Road Salt Contamination: Evidence from the JFK Well, Long Island, NY**
 2. **Frequency-Domain Assessment of Groundwater Response to Maximum and Minimum Salt Winters in a Coastal Aquifer System**
 3. **Wavelet-Based Investigation of Long-Term Aquifer Efficiency Under Repeated Winter Salt Loading**
 4. **Anthropogenic Salinization and Recharge Dynamics: A Spectral Analysis of Groundwater Fluctuations at JFK, Long Island**
 5. **Temporal Shifts in Groundwater Recharge Frequencies Under Chronic Deicing Salt Exposure**
-



Hydrogeology-Focused Titles

6. **Impacts of Winter Deicing Salt on Aquifer Replenishment Dynamics: A Case Study from the JFK Well**
 7. **Chloride Contamination and Groundwater Fluctuation Patterns in a Coastal Aquifer System**
 8. **Evaluating Long-Term Aquifer Efficiency in a High-Salt Urban Environment**
 9. **Groundwater Recharge Response to Freezing Versus Warm Winters in a Salt-Impacted Aquifer**
-



Signal Processing / Spectral Emphasis (Aligns with Your Expertise)

10. **Signal Detection of Salt-Induced Modulation in Aquifer Recharge Dynamics**
 11. **Wavelet and Periodogram Analysis of Groundwater Fluctuations Under Chronic Salinization**
 12. **Spectral Signatures of Aquifer Degradation in a Long-Term Salt-Contaminated System**
 13. **Frequency Evolution of Groundwater Levels in Response to Anthropogenic Winter Forcing**
-



Broader Environmental Impact Angle

14. **Chronic Road Salt Exposure and Its Long-Term Effects on Coastal Aquifer Recharge**
15. **From Short-Term Enhancement to Long-Term Decline: Salt-Induced Changes in Aquifer Dynamics**
16. **Winter Salt Loading as a Driver of Aquifer Efficiency Shifts in Urban Coastal Systems**

If You Want Something Strong, Concise & Memorable

17. **Salt, Signal, and Sustainability: Spectral Evidence of Aquifer Response at JFK**
18. **When Salt Alters the Signal: Detecting Recharge Changes in a Contaminated Aquifer**
19. **Salt and Recharge: A Frequency-Domain Investigation of Aquifer Health**

Review methods:

Review methods:

The methodology is promising, but right now it mixes **what you actually measured** with **what the literature suggests may happen**. That is the main thing to fix.

Here are the main steps:

- 1 **Spatial selection of wells near JFK**
- 2 **Use USGS data only (chloride + groundwater levels)**
- 3 **Classify winters using temperature proxy**
- 4 **Extract DJF groundwater subsets**
- 5 **Compare min-salt vs max-salt using spectral analysis**
- 6 **Interpret differences in periodicities of water level fluctuations**

Our Core Method = DJF Extraction + Winter Classification then spectral analysis:

This is the heart of the methodology. The methodology must explicitly describe the winter-year definition (DJF) and how winters are classified into minimum-salt and maximum-salt using temperature thresholds.

Compare min-salt vs max-salt spectra

Evaluate:

- **power differences**
- **dominant periodicities**
- **amplification or damping**
- **shifts in frequency structure**

About frequency structure:

Spectral analysis is used to compare groundwater behavior between minimum-salt and maximum-salt winters, focusing on differences in spectral power, dominant periodicities, and time-frequency structure. Differences in spectral power or periodic structure between the two winter groups are interpreted as changes in aquifer response under different salt-forcing conditions.

“if cycles decrease or increase then aquifer response changes”. That is exactly what your methodology is designed to detect. Changes in wavelet power or dominant periodicities between minimum-salt and maximum-salt winters are interpreted as differences in aquifer responsiveness, with increased power suggesting stronger or more dynamic groundwater response under higher salt-loading conditions.

About chemistry:

Remove snow chemistry as a primary dataset. If kept, it must be described as background or supporting context, not part of the workflow.

About Soil: Soil / infiltration physics (not part of your method)

You wrote:

- clay compression
- pore clogging
- preferential flow

👉 That is literature discussion, not methodology. Move ALL of that to: Introduction or Discussion NOT Methodology.

Remove or relocate:

- soil physics → Discussion
- snow sampling → remove or clearly label as external/background

So bottom line:

Your methodology is NOT:

- snow chemistry study
- soil process study
- general time series analysis

Your methodology IS:

- 👉 A comparative spectral analysis of groundwater response
- 👉 between low-salt and high-salt winters
- 👉 using USGS well data near JFK

Here are more detailed descriptions but the above should be enough for re-writing the methodology.

Here are my suggestions:

What is strong already

The latest methods version is much better because it:

- narrows the study to a clear spatial target near JFK,
- uses real data sources (snow chemistry + USGS groundwater records) - also needs the map of USGS wells locations,
- separates the idea of **winter salt forcing** from the **groundwater response**, and
- uses spectral tools as an exploratory way to look for seasonal structure.

That is a solid backbone.

The main methodological problems

1) “Min-salt” vs “max-salt” winters are not yet objectively defined

In the code, the winter classes are based on **temperature thresholds**, not on actual road-salt application data. That is fine as a proxy, but the methodology must say that clearly. Right now, the wording makes it sound like you are measuring salt application directly, but you are not. You are really classifying winters by a **freeze/snow proxy** for likely salting intensity.

A stronger definition would be:

- **Min-salt winter** = winter with no qualifying freeze/snow maintenance events, or very few.
- **Max-salt winter** = winter with repeated qualifying events that would likely trigger salting.

2) The snow chemistry and groundwater data are not yet tied together tightly enough

Your abstract suggests a causal chain:
snow salt → infiltration → aquifer salinity.

That is plausible, but the methodology currently does not fully demonstrate that chain.

3) The soil/infiltration discussion is too strong for the methods you actually ran

The code does **not** appear to measure:

- soil texture,
- hydraulic conductivity,
- infiltration rate,
- pore structure,
- or preferential flow.

So the statements about clay compression, pore clogging, and preferential flow are fine as **background literature**, but not as if your data directly tested them.

I would frame those as:

“Previous studies suggest that salt may alter infiltration behavior by changing soil structure and hydraulic properties.”

That is safer and more accurate.

4) Spectral analysis is useful, but it should not be presented as proof of salting effects

Wavelet analysis and periodograms can detect:

- seasonality,
- oscillations,
- multi-year cycles,
- and changes in timing.

Text has not stated how they do by themselves prove road salt is the cause.

So the methodology should say something like:

“Spectral methods were used as exploratory tools to identify periodic structure and potential seasonal forcing in groundwater records.”

That wording is much better than saying the spectra will “show the impact of road salt” directly unless we prove it.

5) Concatenating DJF winters is a methodological weakness that we need to address

The second script correctly notes this caveat, and that is good. The problem is that if you stitch winter seasons together, you create **artificial adjacency** between the end of one winter and the beginning of the next. That can distort:

- periodograms,
- wavelets,
- and any interpretation of cycle length.

What I would change in the methodology language

Instead of saying:

“This study examines how seasonal salt pulses affect groundwater...”

I would say:

“This study evaluates whether winters with stronger freeze/snow maintenance conditions are associated with changes in snow chloride and groundwater conductance/level near JFK.”

That is more precise and matches the code better.

A cleaner methodology paragraph

Here is a version that is closer to what the code actually supports:

Groundwater-level and chloride data were retrieved from the USGS Water Data Services using the R package `dataRetrieval()` (Ref). Wells were selected within a 10 km buffer of John F. Kennedy International Airport (JFK), and records were screened for location validity, sampling history, and continuity. Snow chemistry measurements were used to characterize chloride loading during winter storm events. To examine seasonal groundwater response, daily groundwater-level records were cleaned by removing duplicates and interpolating only short gaps. December observations were assigned to the following hydrologic winter year, and winters were classified using a temperature-based proxy for likely road-salt activity (REF). Groundwater records were analyzed for December–February (DJF) patterns using periodograms and wavelet analysis to detect seasonal forcing, multi-year oscillations, and changes in recharge behavior. Spectral methods were treated as exploratory tools rather than proof of causality, and interpretation was constrained by the irregularity of environmental sampling and the limitations of short-term storm chemistry data.

Bottom line

The methodology is close, but it needs three fixes:

1. make winter classification objective and clearly labeled as a proxy if necessary,
2. soften causal language unless you have direct salt-application data,
3. treat spectral analysis as exploratory and avoid concatenation as the main analytic strategy.

Precipitation JFK

👉 In **GHCN-D** (via **GHCNr**), precipitation (**prcp**) is in:

millimeters (mm)

See below later a better graph

=====

Raw data

=====

```
install.packages("GHCNr") # once
library(GHCNr)
library(dplyr)
```

```
# JFK International Airport station
```

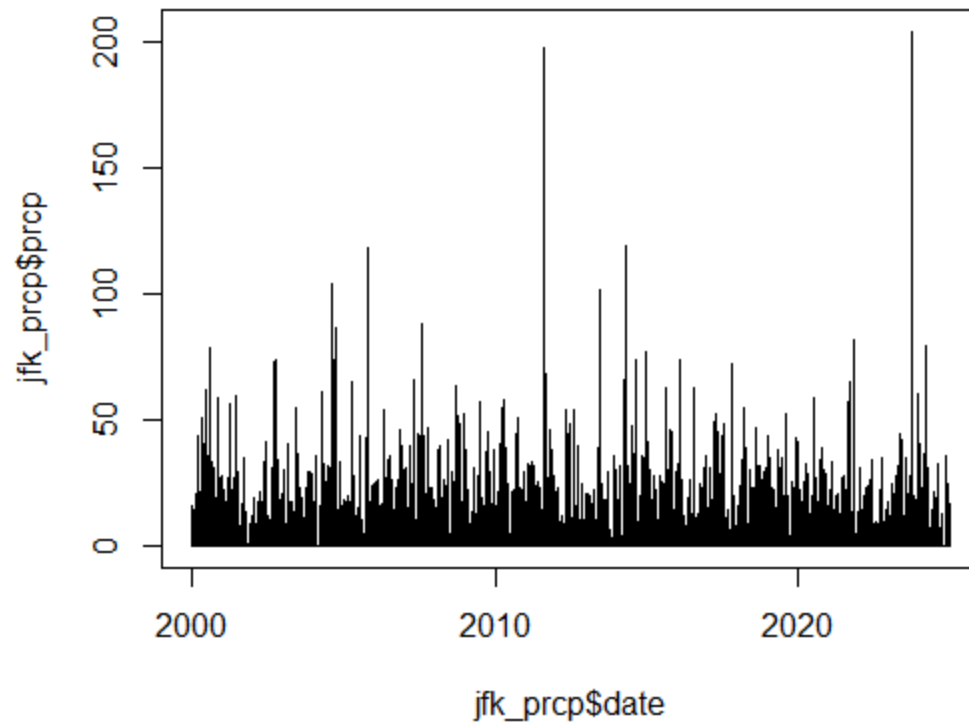
```
jfk_prdp <- daily(
  station_id = "USW00094789",
  start_date = "2000-01-01",
  end_date   = "2024-12-31",
  variables  = "prcp"
)
```

```
plot(jfk_prdp$date, jfk_prdp$prdp, type='l')
```

```
tail(jfk_prdp)
```

```
jfk_prdp <- jfk_prdp %>%
  dplyr::select(date, station_id, prdp)
```

```
print(jfk_prdp)
```



Advanced graph

=====

#####

Here is another better plot of cumulative rain over time:

#####

install.packages("GHCN") # once

install.packages("zoo") # once

install.packages("GHCN") # once

install.packages("dplyr") # once

install.packages("zoo") # once

install.packages("ggplot2") # once

install.packages("GHCN") # once

install.packages("dplyr") # once

install.packages("zoo") # once

install.packages("ggplot2") # once

install.packages("kza") # once


```

library(GHCNr)
library(dplyr)
library(zoo)
library(ggplot2)
library(kza)

# Download daily precipitation
jfk_prdp <- daily(
  station_id = "USW00094789",
  start_date = "2000-01-01",
  end_date = "2024-12-31",
  variables = "prdp"
)

# Clean and prepare
jfk_prdp <- jfk_prdp %>%
  dplyr::select(date, station, prdp, prdp_flag) %>%
  dplyr::arrange(date)

# Replace NA precipitation with 0
jfk_prdp$prdp[is.na(jfk_prdp$prdp)] <- 0

# Moving averages
jfk_prdp <- jfk_prdp %>%
  dplyr::mutate(
    prdp_ma30 = zoo::rollmean(prdp, k = 30, fill = NA, align = "right"),
    prdp_kz = kza::kz(prdp, m = 90, k = 3)
  )

# Plot all together
ggplot2::ggplot(jfk_prdp, ggplot2::aes(x = date)) +
  ggplot2::geom_line(ggplot2::aes(y = prdp), alpha = 0.2) +
  ggplot2::geom_line(ggplot2::aes(y = prdp_ma30), linewidth = 0.7) +
  ggplot2::geom_line(ggplot2::aes(y = prdp_kz), linewidth = 1) +
  ggplot2::labs(
    title = "JFK Precipitation: Raw vs Moving Average vs KZ Filter",
    x = "Date",
    y = "Precipitation (mm)"
  ) +
  ggplot2::theme_minimal()

#####
# BETTER GRAPH BELOW
#####

```

```

# Compute scaling factor
scale_factor <- max(jfk_prdp$prcp, na.rm = TRUE) /
  max(jfk_prdp$prcp_kz, na.rm = TRUE)

ggplot2::ggplot(jfk_prdp, ggplot2::aes(x = date)) +

  # Raw precipitation (left axis)
  ggplot2::geom_line(ggplot2::aes(y = prcp), alpha = 0.25) +

  # KZ smoothed (scaled to match left axis)
  ggplot2::geom_line(
    ggplot2::aes(y = prcp_kz * scale_factor),
    linewidth = 1
  ) +

  ggplot2::scale_y_continuous(
    name = "Daily Precipitation (mm)",

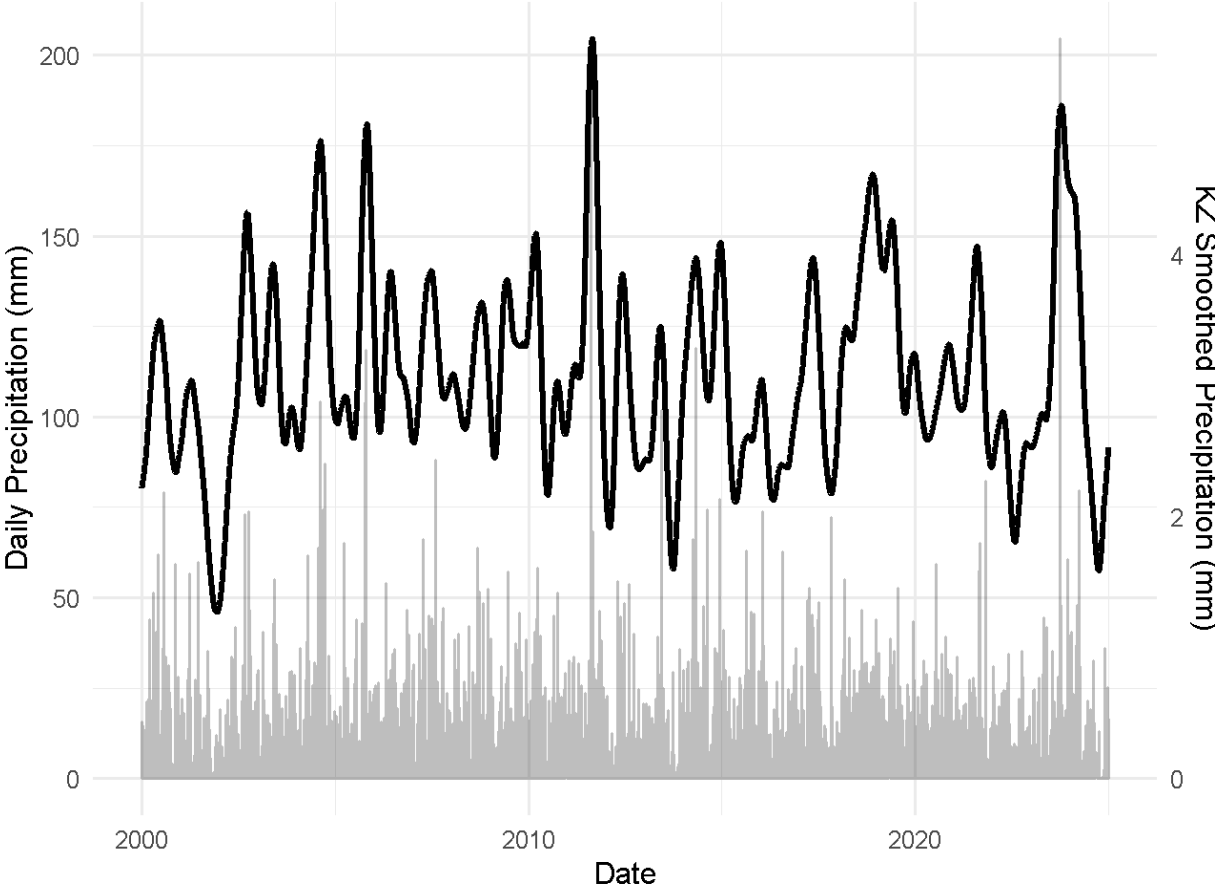
    # Secondary axis (true values of KZ)
    sec.axis = ggplot2::sec_axis(
      ~ . / scale_factor,
      name = "KZ Smoothed Precipitation (mm)"
    )
  ) +

  ggplot2::labs(
    title = "JFK Precipitation with KZ Smoothing (Dual Axis)",
    x = "Date"
  ) +

  ggplot2::theme_minimal()

```

JFK Precipitation with KZ Smoothing (Dual Axis)



Salt in NYC and NY upstate

Road salt pollution affects public health

The Great Flats Aquifer and the Mohawk River in NY have rising levels of sodium and chloride

Road salt pollution continues to receive attention in New York State because of the harm being caused to aquatic ecosystems and municipal water supplies. Several recent high-profile reports warn of the effects of salt on sensitive ecosystems and drinking water in the State. Salt pollution is problematic in the Capital District of NY, which is in the lower part of the Mohawk watershed. My commentary on salt pollution and public health was published on 6 April 2025 in the Sunday Gazette [[here](#) - 1], and it is annotated here with sources and notes:

“Road salt is damaging our drinking water and endangering our health. Recent assessments by the Adirondack Road Salt Reduction task force [2], and the Salinity Management assessment by the NYC Department of Environmental Protection [3] highlight problems to the north and south. However, municipal water quality reports show that salt pollution here in the Capital District is among the most problematic in the State [4].

The Salt Reduction task force report concluded that road salt is harming lakes, streams, and drinking water sources in the Adirondacks [2]. Streams and rivers, groundwater, and ponds have mean chloride levels generally at 10 parts per million (ppm), but higher elsewhere in the state. The report also suggested salt pollution is harming fish and other organisms that rely on freshwater, not saltwater, for their life cycle, and we need to re-evaluate current water quality standards [2].

Drinking water in NYC comes largely from reservoirs in the Catskills, but a recent report by the NYC DEP highlights rising salt pollution in the more urban Croton reservoir [3,5,6]. The City has worked hard to protect its water supply from road salt, with impressive results. The annual water quality report for NYC shows chloride levels averaged 15 ppm in 2024 [7]. However, the assessment warns that some water may

become undrinkable in 83 years [3]. Using this same criteria, some of the water in the Capital District will be undrinkable in the next 15 to 30 years [8].

Drinking water in the Capital District comes from reservoirs, aquifers, rivers, and private wells [i.e. 9,10]. The Mohawk and Hudson corridors are heavily urbanized with significant transportation infrastructure. Noteworthy are salt increases in the Mohawk River and Great Flats Aquifer (GFA), which together serve about a quarter of a million people [11,12,13,14, 15]. Salt contains sodium and chloride, and most municipalities in the Capital District now report levels exceeding the limit for individuals on severely restricted sodium diets [14].

Since road salting began, chloride levels in the Mohawk have risen over 300% and they are elevated year-round [12, 14]. We have spread so much salt across the landscape that we've altered the chemistry of the river. Long-term water quality data from Cohoes and Colonie, which use the Mohawk as a water source, show a sharp rise in sodium and chloride over the last decade [14]. In Cohoes, chloride levels have increased ~125% in that time [14]. The river actually gains sodium and chloride as it passes through the Capital District due to accumulated salt in surrounding sand and gravel deposits [14].

Chloride levels are particularly high in the GFA [12,13,14], one of our most precious resources [9,10]. Chloride in the GFA is typically 200% to 500% higher than the Mohawk [14]. In 2024, Scotia, Rotterdam, and Niskayuna reported levels between 120 and 219 ppm, upper values close to the regulatory limit [14]. Oversalting our roads is part of the problem, and so are large piles of salt stored on top of the aquifer [11].

The trend in salt pollution has been upward for decades [15]. Chloride in Scotia water has risen by nearly 800% since the 1970s, but it has increased 4000% since the

pre-salt days [12]. Chloride promotes pipe corrosion, increasing the risk of lead poisoning [16]. Scotia exceeded the lead action level in 2020, and in 2021, state-mandated tests found lead above the action level in multiple Scotia-Glenville schools [17]. A corrosion-control study was subsequently initiated [18]. The high school had the most problems, with data showing that 45% of samples had high lead [17]. There is no safe level for lead, especially for children, which is why the EPA has set a goal of zero [19].

New York leads the nation in road salt use and salt pollution is a problem across the state [11,12]. We need stricter regulations on salt storage, especially in sensitive areas like the GFA. We also need a strategy to address road salt pollution and part of this will be adopting best management practices [20]. Recent legislation introduced by Senator Harckham would establish a road salt reduction council that may put us on the right path [21]”



Too much salt on a road in Schenectady County (JI Garver).

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Further reading and notes

[01] Garver, JI. 2025. “*Road Salt pollution affects public health*” in Sunday Daily Gazette (Schenectady NY), 6 April 2025 ([here](#)).

[02] Adirondack Road Salt Reduction Task Force Assessment and Recommendations. 2023 ([here](#)).

[03] NYC Department of Environmental Protection, Bureau of Water Supply, “Salinity Management Assessment” 21 March 2025 ([here](#)).

[04] Annual water quality reports are available from municipalities across the Capital District, and elsewhere in the state. For an analysis of these trends in the Capital District, see 11,12,13,14 below.

[05] Gilson, Roger Hannigan, 22 March 2025. “Some New York City water could be undrinkable in 80 years due to upstate road salt: Levels of chloride in water bodies in Westchester and Putnam counties have risen dramatically since the 1990s. Albany Times Union ([here](#)).

[06] Howard, Hillary. 25 March 2025. “Salty Suburban Roads Are Clouding the Future of N.Y.C. Drinking Water: A new environmental report finds that rising salt levels in New York City’s water supply could make some of it undrinkable by the turn of the century.” New York Times ([here](#)).

[07] New York City Drinking Water Supply and Quality report 2024 ([here](#))

[08] Several municipal water sources have salt levels that are increasing dramatically. This is a calculation based on the increase in chloride in the last decade as reported by Annual Water Quality reports for the Village of Scotia. Using both a linear and an exponential fit to the last decade of data (this results in the range given), these are the number of years before chloride equals or surpasses 250 mg/l at the current rate of increase, assuming no other changes.

[09] Winslow, J.D., Stewart, H. G., Jr., Johnston, R. H., and Crain, L. J., 1965, Ground-water resources of eastern Schenectady County, New York, with emphasis on infiltration from the Mohawk River: New York State Conservation Department, Bulletin 57, 148 p.

[10] Brown, AG, 1982 “The Schenectady Area - Chapter 2” in Walker, RM, and Finch, AJ, 1982. Atlas of eleven selected aquifers in New York. UG Geological Survey, Water resources investigations, Open File report 82-553. (Prepared in cooperation with the NY State Department of Health). 255 p. (Brown chapter is p. 15-40).

[11] Garver, JI. 2020, Road Salt: Heavy salt application on roads is damaging the Watershed, In Notes from a Watershed, at Substack.com ([here](#)).

[12] Garver, JI. 2023. Salinization in the Mohawk Watershed: Road Salt contamination in the Great Flats Aquifer threatens public health. In *Notes from a Watershed*, at Substack.com ([here](#))

[13] Garver, JI, Ragland, CJ, Wright, J, Manon, MR, Mundell, H, Smith, JA, 2023. Regional salinization in the lower Mohawk River: effects on urban streams, the Great Flats Aquifer, and raw water for municipal use. In Garver, J.I., Smith, J.A., and Rodak,

C. (eds) Proc. 2023 Mohawk Watershed Symposium, Schenectady, NY, v. 13, p 11-16 ([here](#)).

[14] Garver, JI, and Smith, JA, 2023. Salinization of the Mohawk River and the Great Flats Aquifer in Upstate New York. Geological Society of America Abstracts with programs, v. 55, n. 6. ([here](#)).

[14] Garver, JI, Manon, MR, Rodak, C, and Pillitteri, S., 2025. Sodium and Chloride in the Lower Mohawk River: Implications of imperviousness and urbanization on drinking water sources. In Rodak et al. (eds), 2025 Mohawk Watershed Symposium, Program and abstracts, v. 15, p. 27-32. ([here](#)).

[15] The increase in road salt pollution has been recognized here in the Mohawk Watershed (see 11-14 above), and elsewhere in the Northeast. For examples see: Kaushal, S.S., and other 2022. Freshwater salinization syndrome alters retention and release of chemical cocktails along flowpaths: From stormwater management to urban streams. *Freshwater Science*, 41(3), pp.420-441. AND Dugan, H. A., and Arnott, S. E. (2022). The ecosystem implications of road salt as a pollutant of freshwaters. *WIREs Water*, e1629.

[16] Many studies have shown the link between high chloride, pipe corrosion, and elevated household lead. Hintz and colleagues note that: "Using Cl⁻-contaminated water for drinking water can be dangerous because Cl⁻ results in the dezincification of water plumbing and galvanic corrosion, which increases the thinning of pipe walls and metal leaching" see Hintz, W.D., Fay, L. and Relyea, R.A., 2022. Road salts, human safety, and the rising salinity of our fresh waters. *Frontiers in Ecology and the Environment*, 20(1), pp.22-30 ([here](#)). ALSO see Stets, E.G., Lee, C.J., Lytle, D.A. and Schock, M.R., 2018. Increasing chloride in rivers of the conterminous US and linkages

to potential corrosivity and lead action level exceedances in drinking water. Science of the Total Environment, 613, pp.1498-1509.

[17] See Annual water Quality reports for the Village of Scotia. Lead values in Scotia Schools are presented in a letter from Susan Swartz, superintendent of the Scotia-Glenville school district, dated 11 April 2021. The winter 2021 testing for lead (Pb) in the Scotia-Glenville school district can be seen in spreadsheet form. See also: Supardi, Briana. 12 December 2022. "Scotia residents voice concerns over water lead testing" WRGB. ([here](#)).

[18] See: Prime AE proposal to the Village of Scotia for a Water System Corrosion Control Study (Proposal for Engineering Services) Proposal is dated 3 Nov 2022. Also: Supardi, Briana, 15 Dec 2022. Water quality improvement plan discussed during Scotia Board meeting, at WRGB. Also: see Vaughn-Holdridge, N., 2022. Scotia finds elevated levels of lead in some drinking water samples. The Daily Gazette. 1 December 2022. All the communities evaluated here except Scotia report using corrosion control in their finished water (for most systems this is the addition of orthophosphate to the finished water). Active links for these are at the end of [this post](#), [here](#).

[19] US EPA. For a summary of "Basic Information about Lead in Drinking Water" by the US Environmental Protection Agency see [this website](#). Since the initial round of testing, the action level for lead has been lowered by the NY State.

[20] Since the ADK Salt reduction task force released its report in the Fall of 2023, there has been concern that action on this issue has lagged. See Matson, Zachary, "Advocates plan push for accountability legislation when lawmakers return to Albany" in Adirondack Explorer. 16 September 2024. ([here](#)) and Craig, Gwendolyn, "Settling the

Salt Score: ADKAction holds State accountable for lack of progress. In Adirondack Explorer. 10 February 2025. ([here](#)).

[21] See NY Senate Bill S6976. Establishes the New York road Salt reduction council and the New York Road Salt reduction advisory committee. Status of bill and the bill itself are [here](#). There are a number of other states that have already taken on the issue of road salt and public health. See: “Working together to keep freshwater fresh” Wisconsin Salt Wise ([here](#)) has become a leader in salt solutions, including best management and legal strategies.

temporal

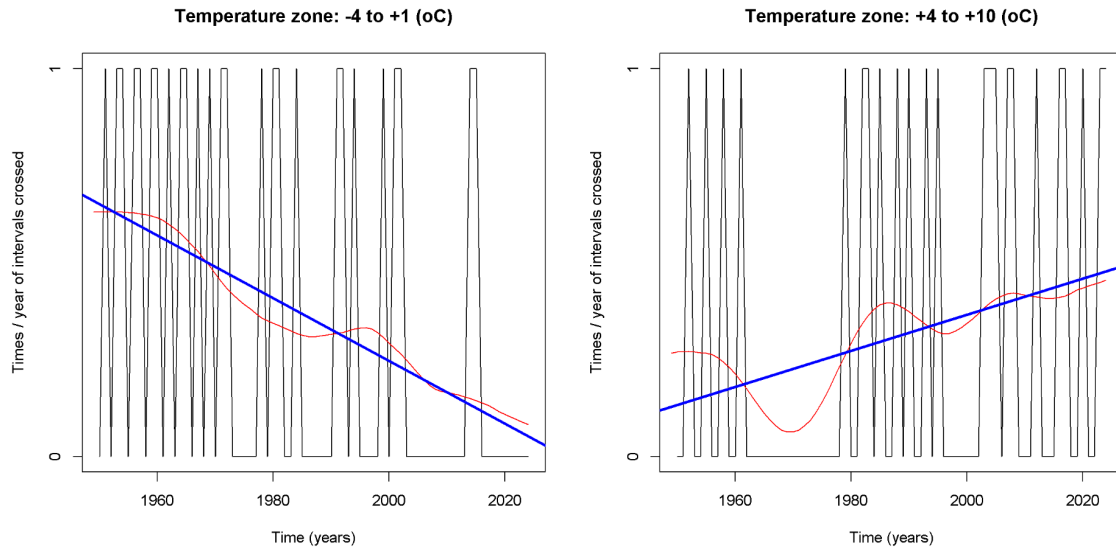


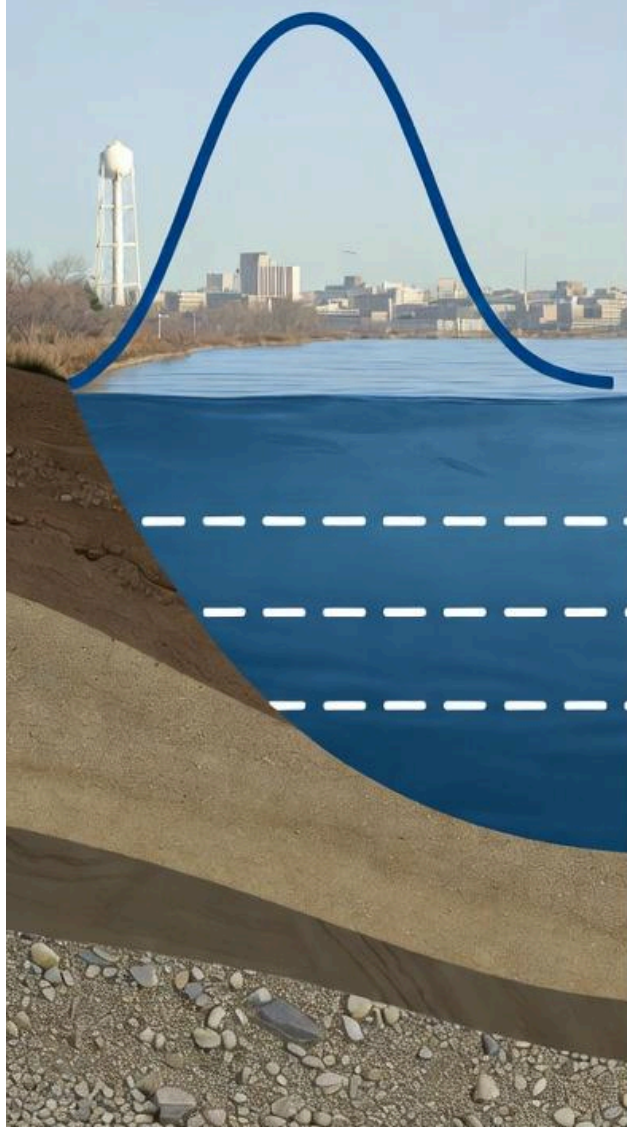
Figure X. Temporal evolution of temperature interval occurrences in New York City (1948–2024), showing the annual frequency of days within two critical temperature ranges: cold conditions (-4°C to $+1^{\circ}\text{C}$; left panel) and warm conditions ($+4^{\circ}\text{C}$ to $+10^{\circ}\text{C}$; right panel). Black vertical lines represent yearly counts of threshold crossings, the red curve indicates the smoothed trend using a Kolmogorov–Zurbenko (KZ) filter, and the blue line represents the linear trend. Results show a decreasing frequency of cold intervals and an increasing frequency of moderate warm intervals over time, reflecting long-term climatic shifts. This analysis builds upon the framework introduced by Gonzalez et al. (2024) in their investigation of climate-driven processes affecting geomorphological dynamics.

Conceptual figure

Low-salt winter:

Stable, seasonal
behavior

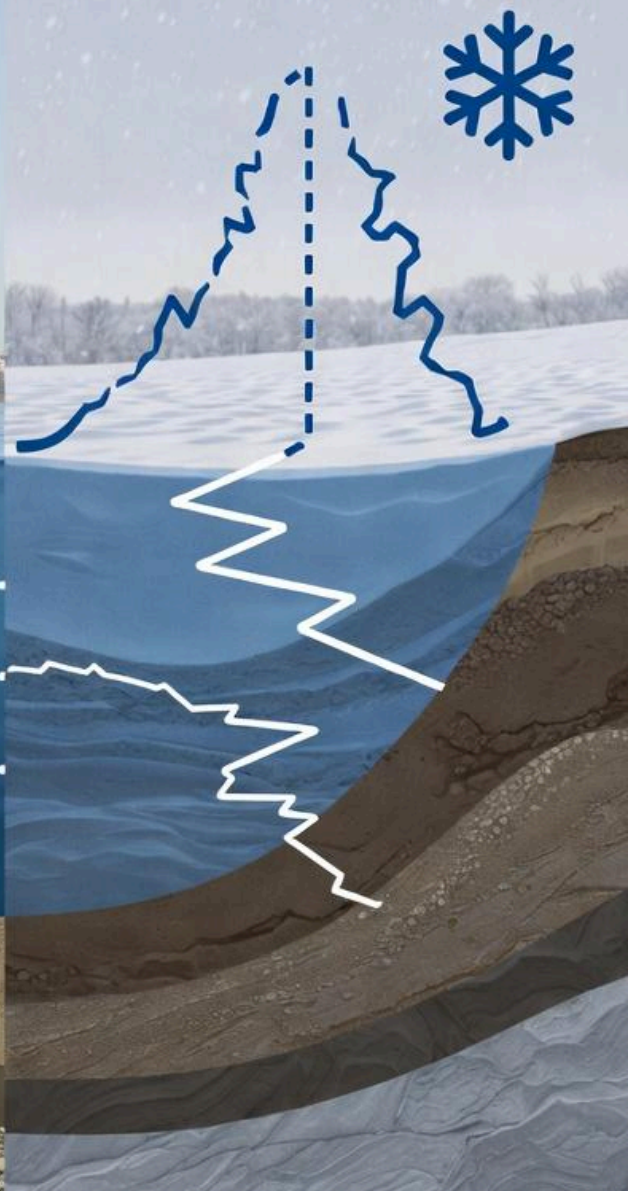
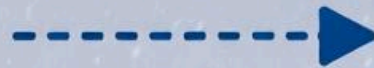
Low-recharge

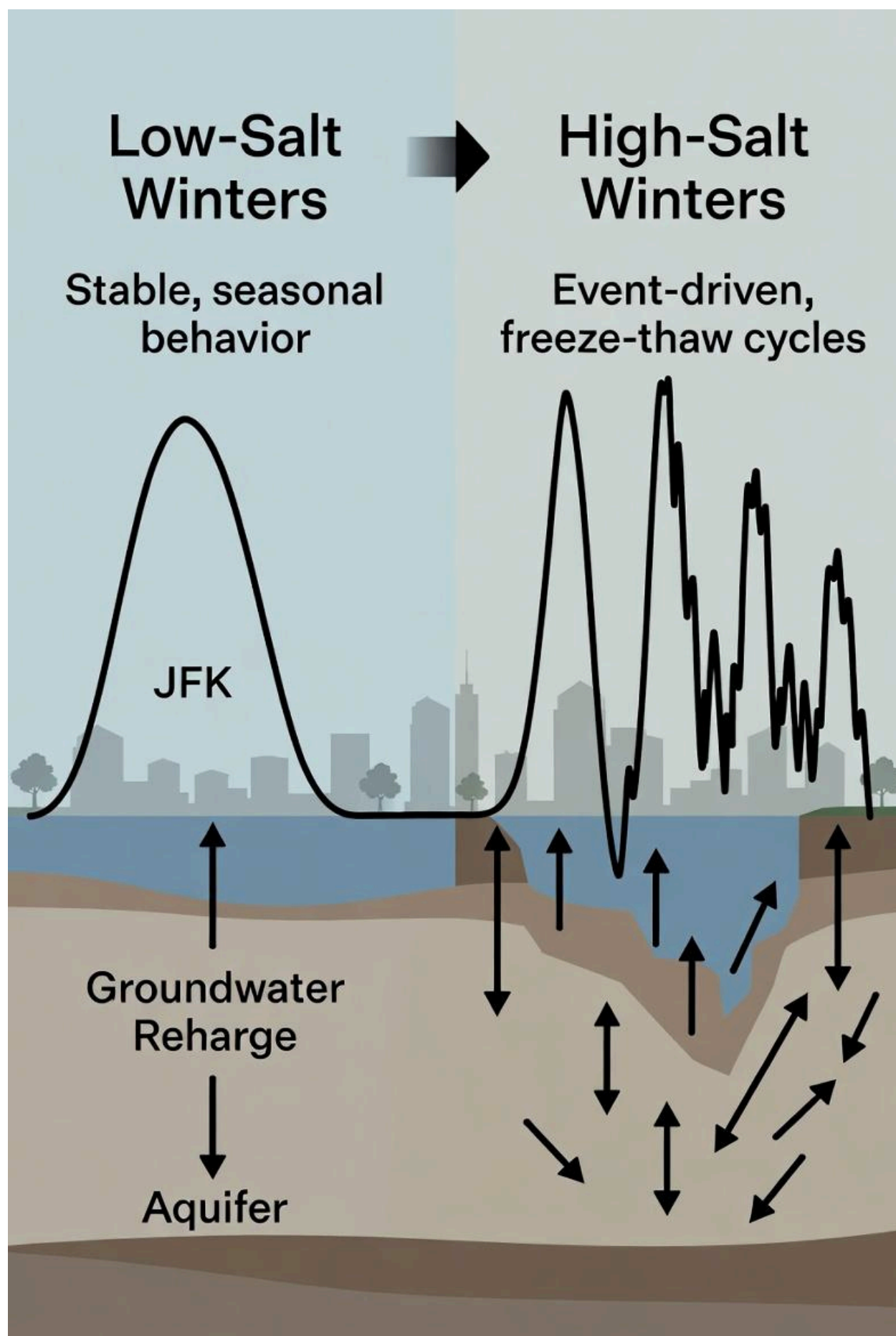


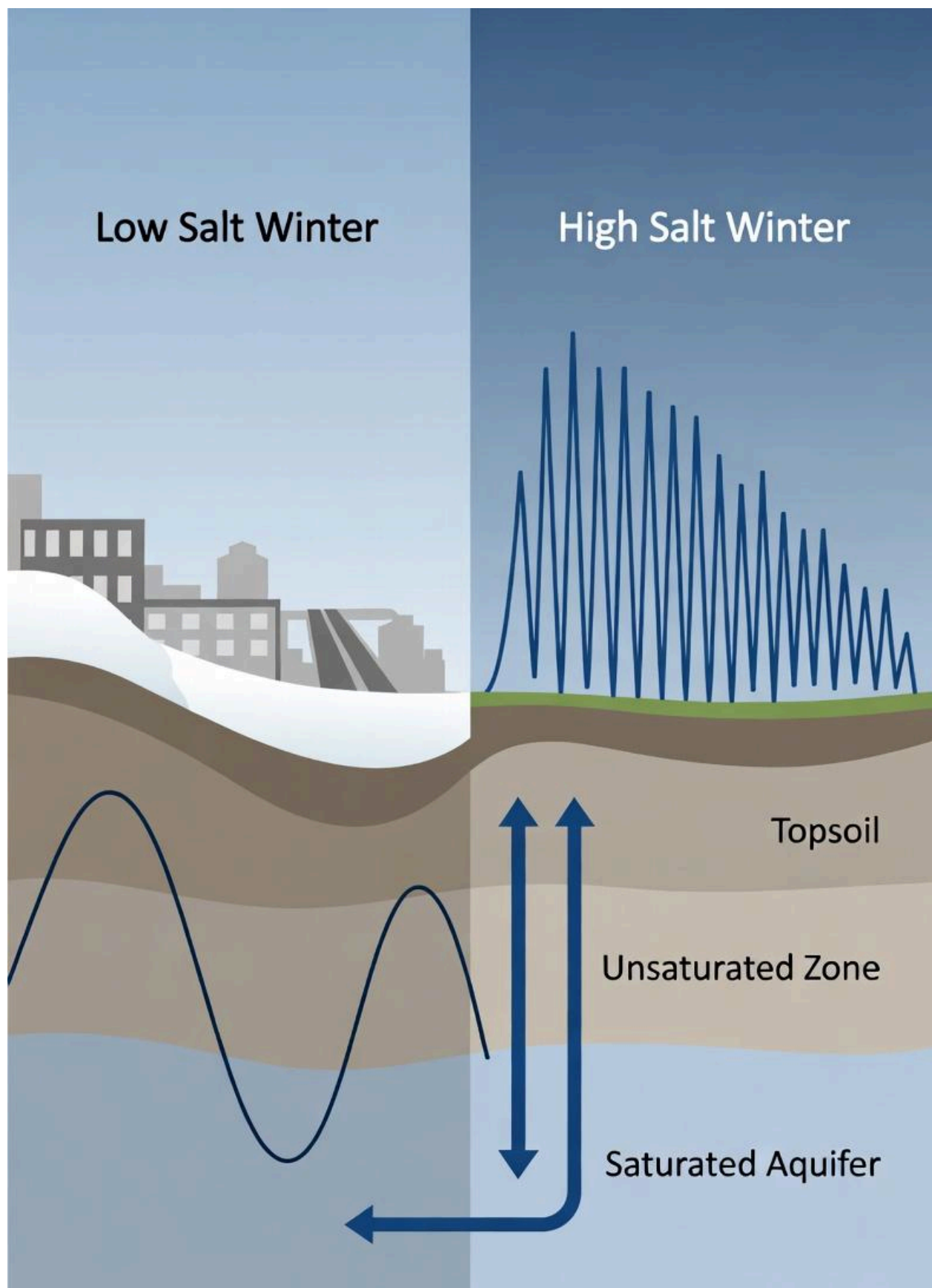
High-salt winter:

Dynamic, event-driven
variability

Variable Recharge







Groundwater Response Shift: Stable Seasonal vs. Dynamic Event-Driven